Hallucination Space Projection for Whisper: Reproduction, an Affine Variant, and a Threshold Baseline

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***Abstract*—**Whisper often outputs fluent text such as "thank you" for audio that contains no speech. A recent paper proposes a training-free fix: remove a low-rank "hallucination subspace" from one decoder layer, only for clips that Whisper already suspects are non-speech. It reports that mean non-speech hallucination rate (HR) falls from 31.31% to 3.74% across three model sizes. We re-implement the method and reproduce it on Whisper small with real data. Our tuned setting lowers held-out HR from 28.41% to 1.27%, and the paper's own setting gives 3.14% against the published 3.07%. However, plain Whisper with a lower no-speech threshold, chosen on the same development data by the same rule, reaches 0.87% HR with lower word error rate (5.93% vs 6.58%) and equal false rejections (1.35%). At equal HR, the projection rejects significantly more speech (McNemar p = 0.024). At the paper's operating point, the edit acts on speech exactly like a threshold. We explain why projecting the last layer fails and test an affine variant that moves states toward the empty-class mean instead of zero. It removes the last-layer failure, but its selected setting generalises poorly (held-out HR 12.61% against 4.0% on development data), and at equal HR it again rejects more speech than a threshold (7 vs 0 utterances, p = 0.016).

***Index Terms*—**speech recognition, Whisper, hallucination, activation editing, voice activity, reproducibility.

I. INTRODUCTION

Large weakly supervised speech recognisers such as Whisper [2] transcribe non-speech audio (silence, noise, music) into plausible text: "thank you", "you", subtitle endings, or a repeated syllable. Because the text is fluent, downstream systems can take it for real speech. Whisper has no separate voice-activity detector. At the first decoding step its decoder predicts a special no-speech token, and the probability of that token, p_{ns}, decides rejection: a segment is dropped when $p_{ns} > \tau = 0.6$. A hallucination occurs when the decoder gives that token a low probability on non-speech audio.

Abbasihafshejani and Jadliwala [1] propose to remove a low-rank subspace from one decoder layer's hidden states, estimated once from non-speech clips that do and do not hallucinate, and to apply the edit only when the unedited p_{ns} already exceeds a small gate γ. Without training they report large HR reductions on three model sizes, at a cost of 0.33 to 4.39 WER points and 0.41% to 9.97% falsely rejected utterances on LibriSpeech.

This report makes four contributions:

- A re-implementation in a single notebook and an end-to-end reproduction on Whisper small, including the paper's own operating point.
- A threshold baseline that the paper omits: plain Whisper with τ chosen on development data by the paper's own selection rule, compared clip by clip with paired significance tests.
- An analysis of what the gate and the edit actually do, and of why editing the last decoder layer fails.
- An affine variant of the projection, which the last-layer analysis motivates.

II. BACKGROUND AND RELATED WORK

Earlier fixes act at other points of the pipeline. External voice-activity detection, as in WhisperX [5], filters the input audio but needs a second model and segmentation. Bag of Hallucinations [3] removes frequent hallucinated phrases from the output, but misses phrases not in its list. Calm-Whisper [4] fine-tunes three decoder heads, which needs training data. Lowering τ acts on the output score and needs nothing at all, but [1] reports that it "changes little" in the range it tests (0.5 to 0.9).

The projection follows activation-editing work on language models, which removes an estimated hallucination subspace from hidden states. Linear concept erasure is known to be optimal when it is affine [6], which motivates our affine variant.

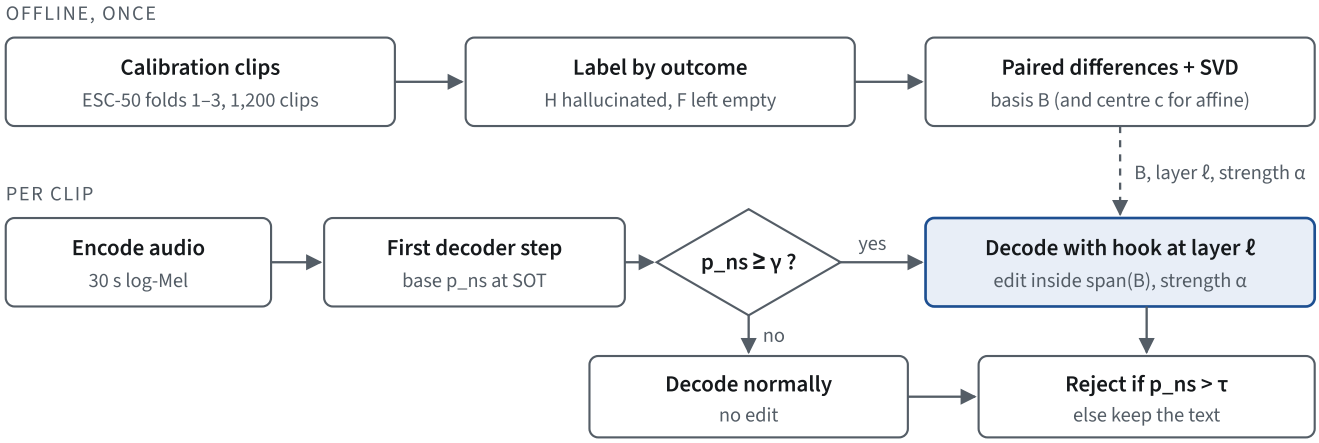


Fig. 1. Gated low-rank decoder projection [1, Alg. 1]. The basis is estimated once; each clip passes the gate before the edit touches it. The only change in our affine variant is the edit itself (Eq. 2).

III. METHOD

A. Calibration sets

Whisper is run on non-speech calibration clips (ESC-50 [7] folds 1–3, 1,200 clips). Each clip goes into one of two groups by its own outcome at $\tau = 0.6$: H, the clips with a non-empty transcript, and F, the clips left empty. For each clip the decoder state at layer ℓ is kept; following the point where p_{ns} is read, we take the state at the start-of-transcript (SOT) position.

B. Hallucination subspace

Rows of H and F are paired and subtracted, so acoustic factors shared by all non-speech audio cancel. The top r right singular vectors of the difference matrix form the basis $B_{\ell,r}$, with orthonormal rows:

$$\Delta_\ell = \begin{bmatrix} h_1^{\text{hall}} - h_1^{\text{empty}} \\ \vdots \\ h_n^{\text{hall}} - h_n^{\text{empty}} \end{bmatrix} = U\Sigma V^\top, \quad B_{\ell,r} = V_{1:r}^\top$$

with $n = \min(N_h, N_f)$ pairs taken in file order.

C. Projection and the affine variant

A forward hook on block ℓ edits every hidden state h, at every position, with strength α:

$$\tilde{h} = h - \alpha\,(hB^\top)\,B \tag{1}$$

With $\alpha = 1$ the component of h inside span(B) is set to zero. Our affine variant instead moves that component to the value of the correctly-empty class, c = mean of the F states at layer ℓ:

$$\tilde{h} = h - \alpha\,((h - c)B^\top)\,B \tag{2}$$

Eq. (2) changes only the target of the edit; B, the gate and the selection procedure are unchanged.

D. Gate and rejection

A first, unedited forward pass gives p_{ns}^{base}. The clip is decoded with the hook only if $p_{ns}^{\text{base}} \geq \gamma$, and the result is rejected if its own $p_{ns} > \tau$. The always-on variant skips the gate. Since every rejected clip has passed the gate, and both decisions read Whisper's own p_{ns}, the natural control is plain Whisper with a lower τ.

IV. EXPERIMENTAL SETUP

A. Data

Calibration uses all 1,200 clips of ESC-50 folds 1–3 (256 hallucinate, 944 stay empty; HR 21.3%). Development data for parameter selection are 400 of these clips and 200 utterances of LibriSpeech [10] dev-clean, as in [1]. Held-out sets are listed in Table I; to fit a laptop GPU we use seeded random subsets of the larger sets.

TABLE I. HELD-OUT EVALUATION SETS

Set	Type	Used	Full
ESC-50 folds 4–5 [7]	non-speech	800	800
UrbanSound8K [8]	non-speech	2,000	8,732
FSD50K eval [9]	non-speech	2,000	8,598
LibriSpeech test-clean	speech	1,000	2,620
LibriSpeech test-other	speech	1,000	2,939

FSD50K: clips labelled Speech or Human_voice removed. Removing music labels as well, as the text of [1] states, leaves only 6,729 clips; the paper's count of 8,621 suggests music was kept, so we keep it.

B. Metrics

Hallucination rate (HR) is the share of non-speech clips with any text left after rejection. Word error rate (WER) is computed on LibriSpeech after Whisper's English normaliser; a rejected utterance counts as all words deleted. False-rejection rate

(FRR) is the share of speech utterances rejected. We report the mean HR over the three non-speech sets and the mean WER and FRR over the two speech sets.

C. Implementation

Everything runs in one notebook on the official `openai-whisper` package: greedy decoding, English forced, no timestamps, audio padded or cut to 30 s. Rejection uses p_{ns} alone, as in Algorithm 1 of [1]. Each clip is decoded once, with or without the hook: p_{ns}^{base} is fixed by the first decoder step, so this equals the paper's "decode, then re-decode if gated" and halves the cost. Weights are cast to fp16 once, decodes are shared between methods, clips that will be rejected are not decoded, and long loops checkpoint and resume. Sanity tests on a tiny random Whisper check before every run that the hook removes exactly the intended component, that the affine edit lands on c, and that the one-pass p_{ns} equals what `whisper.decode` reports. Every per-clip output, including the unfiltered text and p_{ns}, is saved, so other thresholds can be scored without decoding again. The full run took about three hours on an RTX 3050 6 GB laptop GPU (fp16, seed 0, batch 16): about one for Eq. (1) and the baselines, and two for the affine variant.

D. Parameter selection

We follow [1, Sec. 5.1]. Stage 1 sweeps layer $\ell \in \{1, \ldots, 12\}$ and rank $r \in \{1, 2, 4, 8\}$ with always-on projection and $\alpha = 1$. Stage 2 fixes ℓ and r and sweeps $\alpha \in \{0.25, 0.5, 0.75, 1\}$ and $\gamma \in \{0.03, 0.05, 0.10, 0.15\}$ with the gate. In both stages we take the lowest dev HR among settings whose dev WER is within 0.5 points of plain Whisper (ties: lowest WER). The same procedure is run separately for Eq. (1) and Eq. (2). For the threshold baseline the same rule picks τ from a grid of step 0.01.

TABLE II. OPERATING POINTS

Setting	ℓ	r	α	γ
Paper [1], small	10	1	1.0	0.15
Ours, Eq. (1)	11	8	1.0	0.05
Ours, affine Eq. (2)	10	2	1.0	0.05

Plain-Whisper threshold chosen on dev with the same rule: $\tau = 0.17$ ($\tau = 0.14$–0.17 tie at dev HR 0% and unchanged dev WER; we take the largest).

V. RESULTS

A. Parameter selection

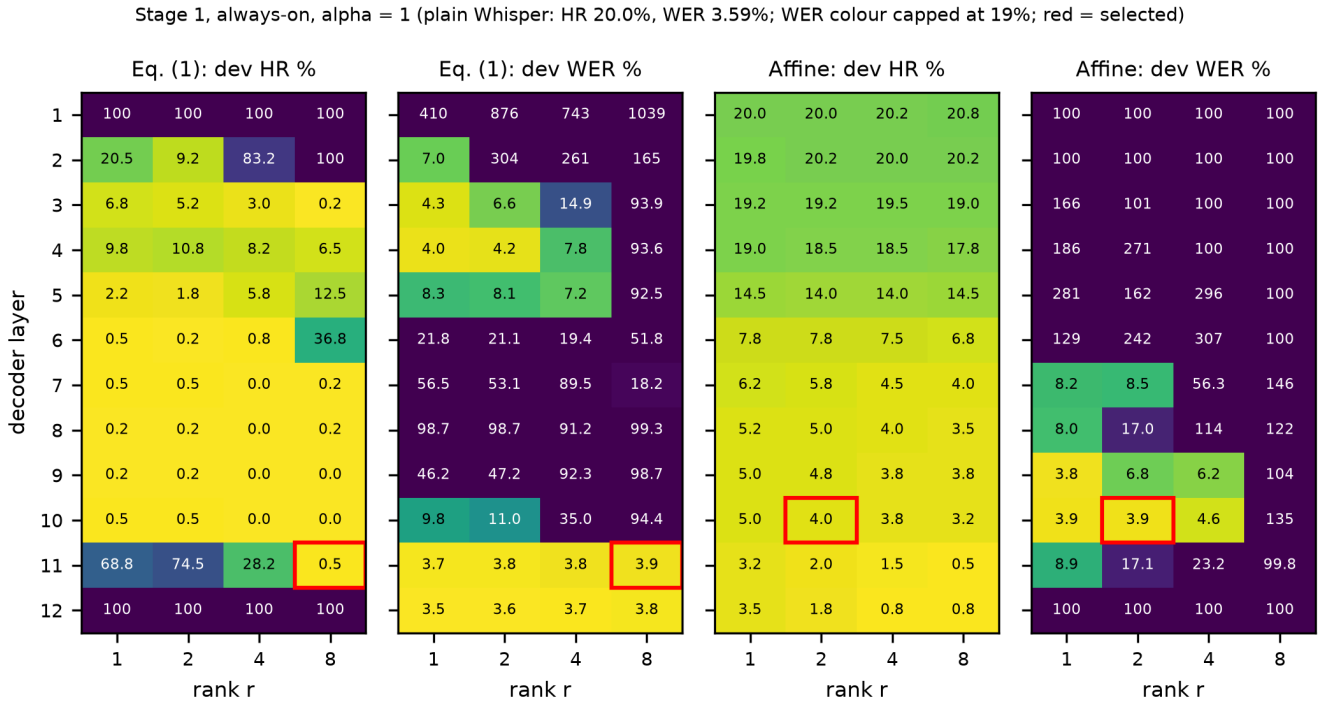


Fig. 2. Stage-1 sweep on development data (always-on, $\alpha = 1$). Left pair: Eq. (1); right pair: affine Eq. (2). Red boxes mark the selected settings. Plain Whisper: HR 20.0%, WER 3.59%.

For Eq. (1) the usable region is narrow (Fig. 2). Nine of 48 settings keep dev WER within 0.5 points: layer 4 at rank 1 and every rank at layers 11 and 12. Of these, only layer 11 at rank 8 also has low HR (0.5%). Middle layers (6–10) remove hallucinations but reject 6.5% to 98.5% of dev speech; the paper's own point, layer 10 at rank 1, gives dev WER 9.8% and FRR 6.5% here. Layers 1 and 12 fail at every rank, with HR 100%. In stage 2, HR falls steadily with α (14.75%, 7.25%, 3.0%, 0.5% for $\alpha = 0.25$ to 1), while γ

changes neither HR nor FRR on dev, because 200 clean utterances rarely reach $p_{ns} \geq 0.03$.

The affine sweep looks different (Fig. 2, right pair; Fig. 4). Dev FRR stays at or below 0.5% at layers 1–11, so the edit no longer rejects speech, and HR falls smoothly with depth instead of jumping between 0% and 100%. Dev WER, however, is 100% or more for most settings: the transcripts are destroyed without being rejected (Sec. V-G). Only three settings stay within 0.5 points (layer 9 at rank 1, layer 10 at ranks 1 and 2), and the rule selects layer 10, rank 2 (dev HR 4.0%, WER 3.89%). In stage 2 HR again falls with α (16.75%, 11.75%, 8.75%, 4.0%), and γ again leaves HR unchanged.

TABLE III. HELD-OUT RESULTS ON WHISPER SMALL (%). HR: LOWER IS BETTER ON NON-SPEECH; WER AND FRR: LOWER IS BETTER ON SPEECH

Method	ESC-50	US8K	FSD50K	Avg HR	WER clean	WER other	FRR clean	FRR other	Avg WER	Avg FRR
Plain Whisper, τ = 0.6	26.12	14.50	44.60	28.41	3.31	7.63	0.0	0.0	5.47	0.00
Bag of Hallucinations [3]	13.75	8.90	18.45	13.70	3.31	7.63	0.0	0.0	5.47	0.00
Always-on, Eq. (1), ℓ11 r8	0.50	1.50	1.75	1.25	3.37	10.25	0.2	3.3	6.81	1.75
Gated, Eq. (1), ℓ11 r8	0.50	1.50	1.80	1.27	3.50	9.66	0.2	2.5	6.58	1.35
Always-on, paper point	1.38	0.55	7.10	3.01	8.20	19.49	6.7	14.7	13.84	10.70
Gated, paper point	1.38	0.65	7.40	3.14	3.91	8.41	1.4	2.4	6.16	1.90
Gate only (reject if $p_{ns} \geq 0.05$)	0.00	0.00	0.20	0.07	13.07	20.68	14.9	20.7	16.87	17.80
Always-on, affine, ℓ10 r2	8.38	6.60	22.85	12.61	2.92	7.87	0.1	0.6	5.40	0.35
Gated, affine, ℓ10 r2	8.38	6.60	22.85	12.61	3.32	8.03	0.1	0.6	5.68	0.35
Gated, affine, ℓ11 r8	1.00	1.30	1.80	1.37	13.03	20.65	0.9	5.4	16.84	3.15
Gated, affine, paper point	9.25	7.00	24.25	13.50	3.31	7.65	0.1	0.1	5.48	0.10
Plain Whisper, τ = 0.17 (dev)	0.25	0.30	2.05	0.87	3.72	8.14	1.2	1.5	5.93	1.35

Paper [1], small, gated: HR 1.53 / 0.84 / 6.84; WER 4.37 / 9.13; FRR 0.41 / 2.58. Plain: HR 23.50 / 12.33 / 21.35; WER 4.04 / 8.38.

B. Held-out results and agreement with the paper

Gated projection at the paper's operating point reproduces the published HR closely (1.38 / 0.65 / 7.40 vs 1.53 / 0.84 / 6.84). The WER increase over plain Whisper is similar (+0.60 / +0.78 points vs +0.33 / +0.75), and so is test-other FRR (2.4% vs 2.58%); test-clean FRR is higher (1.4% vs 0.41%). Always-on projection at the paper's point costs much more speech than reported (WER 8.20 / 19.49 vs 4.51 / 11.48). Plain Whisper's FSD50K HR is 44.6% here against 21.35% in [1]; the paper prints 21.35% for both small and large-v3, and our value is close to its medium value (41.95%). Music does not explain the gap: plain Whisper hallucinates on 38.8% of our FSD50K clips with a music label and on 46.2% of the rest. Our tuned Eq. (1) setting lowers mean HR to 1.27%, below the paper's point (3.14%), at slightly higher WER (6.58% vs 6.16%) and lower FRR (1.35% vs 1.90%).

C. A tuned threshold matches or beats the projection

On development data the threshold alone already beats the projection: τ = 0.17 gives dev HR 0.00% with dev WER unchanged (3.59%), against 0.50% and 3.57% for the selected Eq. (1) setting. On held-out data τ = 0.17 gives HR 0.87% vs 1.27%, WER 5.93% vs 6.58%, and the same FRR (Table III); it is lower or equal on all three. Within the dev tie the choice matters little: τ = 0.15 gives HR 0.71%, WER 6.18%, FRR 2.0%. No projection setting lies below or to the left of the threshold curve (Fig. 3). The no-speech probability already separates speech from non-speech almost perfectly (AUROC 0.9991), and projection does not improve that (0.9986 gated, 0.9983 always-on).

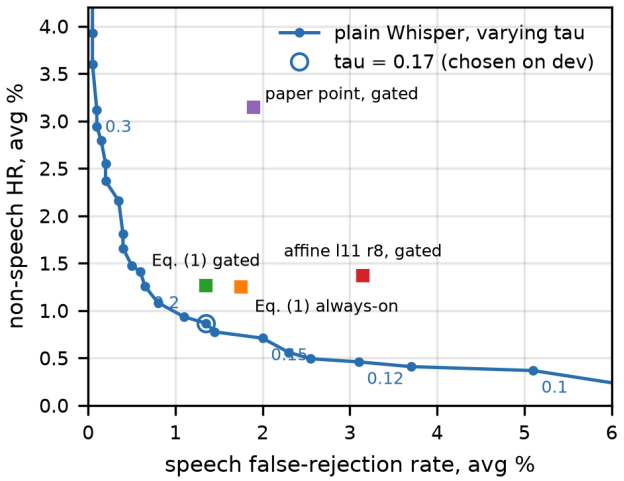


Fig. 3. Held-out trade-off. Line: plain Whisper as τ varies (no projection); the ring marks τ chosen on dev. Squares: projection settings. Points off the axes are listed in Table III.

D. Paired significance tests

Both methods run on the same clips, so we compare them clip by clip: an exact McNemar test [11] on the clips that only one method gets wrong, and a paired bootstrap over utterances (2,000 resamples) for WER. Table IV compares each projection with plain Whisper at the τ that gives the same held-out HR.

TABLE IV. PAIRED TESTS AT EQUAL HR

Comparison	Only proj.	Only τ	p
Gated Eq. (1) ℓ11 r8 vs τ = 0.2			
Speech rejected	24	10	0.024
Non-speech hallucinated	45	46	1.0
Avg WER difference		+0.95 [0.40, 1.56]	
Gated, paper point vs τ = 0.3			
Speech rejected	36	0	$3{\cdot}10^{-11}$
Non-speech hallucinated	91	91	1.0
Avg WER difference		+0.68 [0.40, 0.99]	
Gated affine ℓ10 r2 vs τ = 0.46			
Speech rejected	7	0	0.016
Non-speech hallucinated	192	169	0.25
Avg WER difference		+0.21 [0.06, 0.41]	

"Only proj." / "only τ": clips that only one method gets wrong. WER difference: projection minus threshold, in points, with 95% interval.

At equal HR every projection rejects significantly more speech than the threshold, and its WER is significantly higher, by 0.95, 0.68 and 0.21 points. On non-speech no projection differs significantly from the threshold that matches its HR.

E. What the gate and the edit do

At the paper's operating point (γ = 0.15) the gate opens for 14 of 1,000 test-clean and 26 of 1,000 test-other utterances, and the projection then rejects 14 of 14 and 24 of 26 of them. On speech, the method is therefore the threshold "reject if $p_{ns} \geq 0.15$". Plain Whisper at τ = 0.15 gives the same speech numbers (WER 3.91 / 8.45, FRR 1.4 / 2.6, against 3.91 / 8.41 and 1.4 / 2.4) with lower HR (0.71% vs 3.14%).

Our setting differs. With γ = 0.05 the gate opens for 99.9% of non-speech clips, so on non-speech the gated method is effectively always-on. On speech it opens for 149 and 207 utterances and keeps all but 2 and 25 of them, which is why rejecting every gated clip (gate only, Table III) is a poor control. The edit does not harm the transcripts it keeps: on the same projected and accepted utterances, WER goes from 4.55% to 4.65% (test-clean, n = 147) and from 10.80% to 10.49% (test-other, n = 182); for always-on, from 3.30% to 3.18% and from 7.73% to 7.76%. The whole WER cost is rejection, and the rejected utterances are short (at the paper's point 9.1 and 7.4 words against 20.2 and 18.2 on average).

F. Why the last layer fails

The state after the last block goes directly to the final LayerNorm and output embedding, so p_{ns} can be recomputed from the saved calibration states, with and without the edit (the recomputation matches the saved values within 0.004). Because calibration labels are defined by p_{ns}, the direction separating H from F is largely the no-speech read-out: the coordinate on the first basis vector correlates with log p_{ns} at −0.73. That coordinate is far from zero in both groups (−79.6 for hallucinated clips, −91.6 for empty ones). Eq. (1) sets it to zero, moving every clip 80 to 90 units in the direction that lowers p_{ns}: the share of calibration clips with $p_{ns} > 0.6$ drops from 78.7% to 0%, at every rank. Nothing is rejected any more, which is the 100% HR of Fig. 2. Eq. (1) is therefore not a neutral removal; it moves states to the origin of the subspace, far from where either class lies. The affine edit (Eq. 2) moves the coordinate to the F mean instead, and does what the method intends: the share of calibration clips with $p_{ns} > 0.6$ rises from 78.7% to 95.2% at rank 1 and to 99.4% at rank 8. At layer 12 dev HR is now 3.5% to 0.75% rather than 100%. It is still not usable there, because every speech utterance is either rejected (dev FRR 57.5% at rank 1, 5.5% at rank 8) or decoded into garbage.

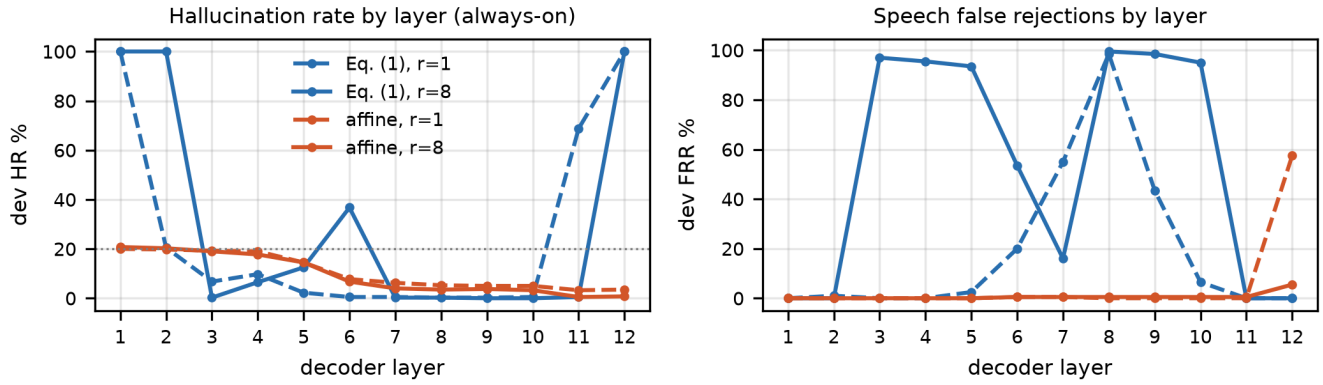


Fig. 4. Development HR (left) and FRR (right) by decoder layer, always-on, α = 1, for Eq. (1) (blue) and the affine Eq. (2) (orange) at ranks 1 and 8. Dotted: plain Whisper.

G. Affine variant

Held-out results. The selected affine setting (ℓ10 r2, α = 1, γ = 0.05) keeps almost all speech: FRR is 0.35% and WER 5.68%, against 1.35% and 6.58% for Eq. (1). It removes far fewer hallucinations, though: mean HR is 12.61%, against 4.0% on development data and 1.27% for Eq. (1) (Table III). The gap comes mostly from FSD50K (22.85%), whose sound classes are not in ESC-50; even ESC-50 folds 4–5 reach 8.38%. The tuned Eq. (1) setting moved much less from dev to test (0.5% to 1.27%). At the paper's point the affine edit gives similar numbers (HR 13.50%, FRR 0.10%).

Against the threshold. τ = 0.17 is better on hallucinations (0.87% vs 12.61% HR) but rejects more speech (24 vs 4 utterances, $p = 1.8 \cdot 10^{-4}$); the WER difference is not significant (−0.25 [−0.55, 0.03] points). At equal HR (τ = 0.46) the threshold again dominates: it rejects no speech where the affine method rejects 7 (Table IV). On Fig. 3 the selected affine points lie above the plotted range, and affine at ℓ11 r8 lies to the right of the threshold curve, like the other projections.

Why the edit fails at other settings. The centre c is the mean state at the SOT position, but the hook edits every position. At later positions the states lie far from c, so the edit adds a large constant shift. On gated speech at ℓ11 r8 the decoder then emits only language tokens ("<|zh|><|zh|>..."): HR is low (1.37%), but WER on the edited utterances is 99.6%, so mean WER is 16.84% (Table III). This is the 100% WER at FRR 0% in Fig. 2. Estimating c per position, or editing only the SOT position, would be the natural next step.

Side effect on speech. Always-on affine lowers test-clean WER from 3.31% to 2.92% (paired bootstrap −0.39 [−0.77, −0.09] points). Most of the gain comes from a few utterances that plain Whisper cut short, and it does not carry over to test-other (+0.24 [−0.17, 0.69]), so we do not count it as an improvement.

VI. DISCUSSION

Our results point to seven issues in the evaluation of [1].

1. *The key control is missing.* The τ sweep in [1] stops at 0.5 and is described as changing little, yet its own medium ablation shows plain HR falling from 29.49% to 13.08% as τ goes from 0.6 to 0.4. A threshold chosen by the paper's own rule matches or beats every projection setting we tried.
2. *At the published small setting the edit adds nothing on speech.* Every gated speech utterance is rejected; the gate-open rate on speech and the fate of projected utterances should be reported.
3. *WER mixes two errors.* Counting a rejected utterance as all deletions is correct but hides whether the edit damages transcription. On the same accepted utterances it does not; the cost is rejection only. The paper's own numbers imply a rejected-word share of 0.32 to 0.88 times FRR, i.e. short utterances are rejected.
4. *Projection to zero is not neutral* (Sec. V-F). An affine target, the principled choice [6], fixes the last layer, but a single centre applied at every position creates new failures (Sec. V-G).
5. *Calibration labels are defined by p_{ns},* so the subspace partly re-learns the no-speech read-out, consistent with unchanged AUROC.
6. *Tuning ignores noisy speech.* Our Eq. (1) setting had 0% FRR on dev-clean but 2.5–3.3% on test-other.
7. *Internal inconsistencies.* The validation WER in Fig. 1 and Sec. 5.1 of [1] differ (3.94% vs 11.59%); the medium α is given as 0.75 and 1.0; the FSD50K plain HR is identical (21.35%) for small and large-v3; Table 5 at τ = 0.6 disagrees with Table 1 for medium (plain HR 29.49% vs 27.66%, projected 5.09% vs 4.06%); the FSD50K count fits removing speech but not music; and Calm-Whisper is compared under a different HR definition.

Wilson intervals on the paper's own large-v3 rates also qualify one claim: WhisperX VAD beats gated projection on ESC-50 (4.13% [2.96, 5.74] vs 8.38% [6.65, 10.50]), ties on UrbanSound8K, and loses only on FSD50K (8.55% vs 1.15%).

VII. LIMITATIONS

We test Whisper small only, with one seed and held-out subsets of 1,000–2,000 clips per set; medium and large-v3, where [1] reports its largest gains, need a larger GPU. Development data are small (400 non-speech and 200 speech clips), so dev HR moves in steps of 0.25% and several thresholds tie. We did not run the WhisperX VAD baseline. fp16 results depend slightly on batch size, which we kept fixed at 16.

VIII. CONCLUSION

Gated hallucination space projection reproduces on Whisper small: it removes most non-speech hallucinations, our rates at the paper's setting are within about a point of the published ones, and the edit does not harm the transcripts it keeps. Its benefit, however, comes from Whisper's own no-speech probability. Lowering the rejection threshold, chosen on development data by the same rule, does at least as well with no change to the model, and at the paper's operating point the method is that threshold. Moving states to the empty-class mean instead of zero fixes the last-layer failure and nearly removes false rejections, but its single SOT-position centre breaks decoding at many settings. Its selected setting also generalises poorly to held-out sounds, so it does not beat the threshold either. Before projection can be called an improvement, the comparison against a tuned threshold should be repeated on medium and large-v3.

AUTHOR CONTRIBUTIONS

Nongmaithem Hans researched the topic and selected the paper. Hans ran all experiments on a personal laptop GPU (calibration, parameter sweeps and held-out evaluation, the method's equivalent of training) and optimised the pipeline to fit a 6 GB GPU: fp16 weights, shared decoding, skipping clips that are rejected anyway, and checkpointing with resume. Hans designed and carried out the evaluations and comparisons: the comparison with the paper, the threshold baseline, the paired significance tests, the gate and last-layer analyses, and the affine variant. Hans also prepared the presentation from this report.

Ravikant Sharma downloaded and organised the four datasets (ESC-50, UrbanSound8K, FSD50K and LibriSpeech) and wrote the initial code of the notebook. Ravikant wrote this report from the results of the runs.

Both authors discussed the findings and reviewed the final report.

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